

EXPERIMENT: 10

A PYTHON PROGRAM TO IMPLEMENT DIMENSIONALITY REDUCTION – PCA

Aim:

To implement dimensionality reduction using Principal Component Analysis (PCA).

Procedure:

1. Import required libraries.
2. Load dataset (Iris dataset).
3. Apply PCA to reduce dimensions.
4. Transform data into principal components.
5. Visualize reduced data.

CODE:

```
from sklearn.datasets import load_iris

from sklearn.decomposition import PCA

import matplotlib.pyplot as plt

data = load_iris()

X = data.data

y = data.target

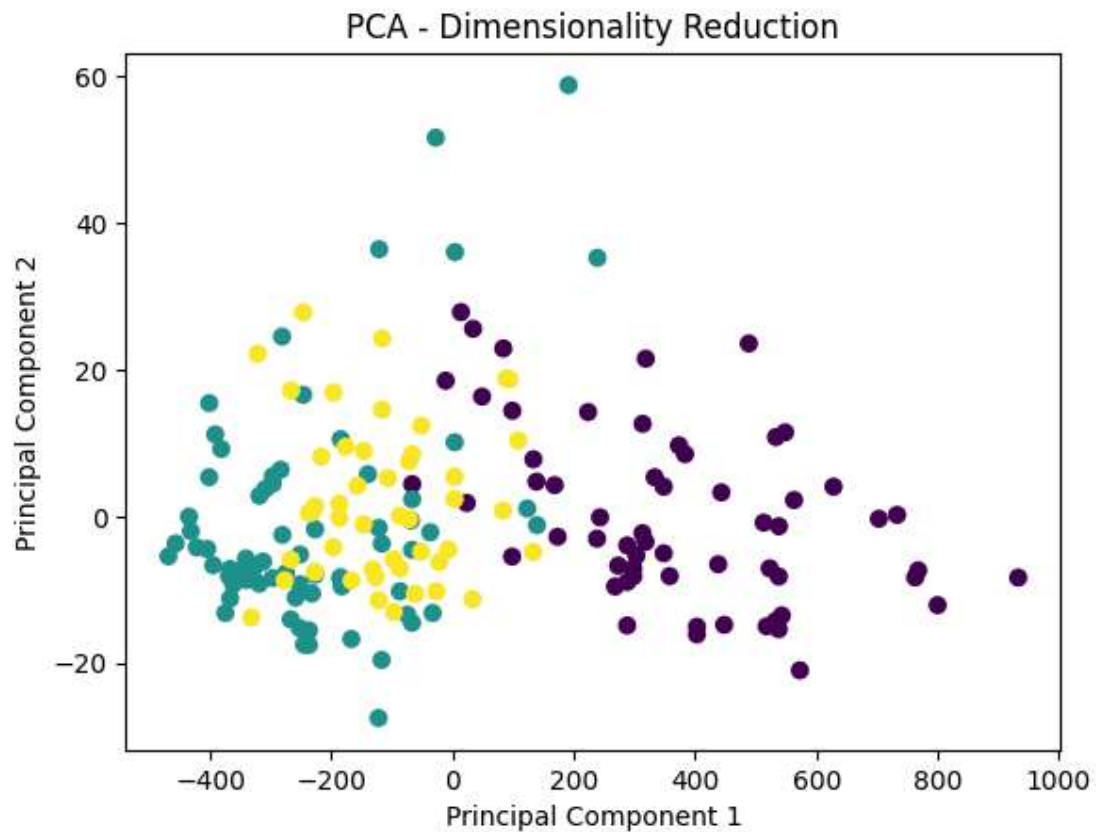
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X)

plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
```

```
plt.xlabel("Principal Component 1")  
plt.ylabel("Principal Component 2")  
plt.title("PCA - Dimensionality Reduction")  
plt.show()
```

OUTPUT:



KNN Accuracy: 0.8055555555555556

RESULT:

PCA successfully reduced the dataset dimensions while preserving important information.